

We use $P(\cdot)$ (sometimes also written $\Pr(\cdot)$) to refer to a probability function or probability distribution. When using an upper letter (e.g., X or $Cavity$), we refer to the random variable; when using lowercases we refer to a specific value of the corresponding random variable. So, $P(Cavity)$ is a probability distribution over variable $Cavity$; whereas $P(cavity)$ is a shorthand for $P(Cavity = true)$ and $P(\neg cavity)$ is a shorthand for $P(Cavity = false)$.

PART 1: Probabilities

- (a) Prove that if X is a random variable over a finite range R , then $P(\bigvee_{x \in R} X = x) = 1$
- (b) Would it be rational for an agent to hold the three beliefs $P(A) = 0.4$, $P(B) = 0.3$, and $P(A \vee B) = 0.5$? If so, what range of probabilities would be rational for the agent to hold for $A \wedge B$? Make up a table like Figure 13.2 or [this one](#), and show how it supports your argument about rationality. Then draw another version of the table where $P(A \vee B) = 0.7$. Explain why it is rational to have this probability, even though the table shows one case that is a loss and three that just break even. (Hint: what is Agent 1 committed to about the probability of each of the four cases, especially the case that is a loss?)
- (c) Consider the domain of dealing 5-card poker hands from a standard deck of 52 cards, under the assumption that the dealer is fair.
 - i. How many atomic events are there in the joint probability distribution (i.e., how many 5-card hands are there)?
 - ii. What is the probability of each atomic event?
 - iii. What is the probability of being dealt a royal straight flush (the ace, king, queen, jack and ten of the same suit)?
 - iv. What is the probability of being dealt four-of-a-kind (i.e., four cards of different suit but same face value)?

PART 2: Conditional probability

- (a) Prove, formally, that $P(A \mid B \wedge A) = 1$.
- (b) Consider again the domain of dealing 5-card poker hands from a standard deck of 52 cards, under the assumption that the dealer is fair. You are told that the probability drawing two cards from a deck of 52 and them both being an ace is $\frac{1}{221}$. You take the deck and draw the first card — it is the ace of spades! What is the probability that the second card you draw will also be an ace? (even though this is easy to work out using binomials, use conditional probability for this question)
- (c) Given the full joint distribution shown in the table below, calculate the following:

	<i>toothache</i>		$\neg$ <i>toothache</i>	
	<i>catch</i>	$\neg$ <i>catch</i>	<i>catch</i>	$\neg$ <i>catch</i>
<i>cavity</i>	.108	.012	.072	.008
$\neg$ <i>cavity</i>	.016	.064	.144	.576

Note that $P(\text{Cavity})$ denotes a vector of values for the probabilities of each individual state of Cavity. Also note here we use the uppercase (e.g., Cavity) to denote a variable, whereas lowercase (e.g., cavity) to denote a constant.

- i. $P(\text{toothache})$
 - ii. $P(\text{Cavity})$
 - iii. $P(\text{Toothache} \mid \text{cavity})$
 - iv. $P(\text{Cavity} \mid \text{toothache} \vee \text{catch})$
- (d) After your yearly checkup, the doctor has bad news and good news. The bad news is that you tested positive for serious disease and that the test is 99% accurate (i.e., the probability of testing positive when you do have the disease is 0.99, as is the probability of testing negative when you don't have the disease). The good news is that this is a rare disease, striking only 1 in 10,000 people of your age. Why is it good news that the disease is rare? What are the chances that you actually have the disease?
- (e) Prove, formally, that $P(A \wedge B \wedge C) = P(A \mid B \wedge C) \times P(B \mid C) \times P(C)$.
- (f) Prove Bayes' Theorem: $P(A \mid B) = \frac{P(B \mid A) \times P(A)}{P(B)}$.
- (g) For each of the following statements, either prove it is true or give a counterexample:
- i. If $P(a \mid b, c) = P(b \mid a, c)$, then $P(a \mid c) = P(b \mid c)$
 - ii. If $P(a \mid b, c) = P(a)$, then $P(b \mid c) = P(b)$
 - iii. If $P(a \mid b) = P(a)$, then $P(a \mid b, c) = P(a \mid c)$

End of tutorial worksheet